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Studierichting: Informatica
Oefeningenreeks 1D nr. 19.1

$$19.1) \ z^3 + 11z^2 + 23z - 35$$

$$\begin{array}{c|cccc} & 1 & 11 & 23 & -35 \\ 1 & & 1 & 12 & 35 \\ \hline z_1 = & 1 & 12 & 35 & 0 \end{array}$$
$$(z - 1)(z^2 + 12z + 35)$$

$$\Rightarrow D = 144 - 4 \cdot 1 \cdot 35 = 144 - 140 = 4$$

$$\Rightarrow \sqrt{D} = 2$$

$$z_2 = \frac{-12 + 2}{2} = -5$$

$$z_3 = \frac{-12 - 2}{2} = -7$$

$$\Rightarrow (z - 1)(z + 5)(z + 7)$$

References